

KAWAGUCHIKO, Japan

WMO: 476400

Lat: 35.5006N

Lon: 138.7608E

Elev: 861

StdP: 91.40

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-7.1	-18.3	0.8	-2.0	-16.8	0.9	-1.6	7.2	-0.2	6.4	0.0	1.3	240	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	30.3	22.4	29.0	21.9	27.6	21.4	23.3	28.2	22.7	27.2	22.1	26.1	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.5	25.7	21.4	17.8	25.1	20.8	17.2	24.5	74.4	28.4	71.9	27.3	69.4	26.3	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.9	5.1	4.4	-11.9	32.8	1.5	0.9	-13.0	33.5	-13.9	34.0	-14.7	34.6	-15.8	35.2
			-12.8	24.6	1.4	0.8	-13.8	25.2	-14.7	25.7	-15.5	26.1	-16.5	26.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-0.2	0.8	4.4	9.7	14.7	18.2	22.5	23.2	19.2	13.6	7.9	2.7
	DBStd	8.58	2.70	3.31	3.50	3.71	2.78	2.71	2.51	2.18	3.06	3.12	3.25	3.04
	HDD10.0	1114	317	259	176	49	2	0	0	0	0	4	81	227
	HDD18.3	2887	576	492	431	258	116	35	2	2	27	152	313	484
	CDD10.0	1646	0	1	4	41	148	244	388	411	277	115	17	1
	CDD18.3	377	0	0	0	0	4	30	132	154	53	4	0	0
	CDH23.3	2018	0	0	0	5	42	133	743	864	215	14	0	0
	CDH26.7	497	0	0	0	1	3	19	209	232	32	1	0	0
Wind	WSAvg	1.9	1.9	2.0	2.2	2.2	2.0	1.8	1.8	1.8	1.7	1.7	1.7	1.9
Precipitation	PrecAvg	1444	54	53	92	100	120	153	159	175	234	174	81	49
	PrecMax	2108	153	116	170	230	278	270	317	414	510	661	247	130
	PrecMin	866	2	3	47	31	50	69	54	35	54	21	3	4
	PrecStd	262	40	29	34	48	54	56	77	100	127	130	53	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	14.7	18.7	23.7	26.7	28.8	32.2	31.9	29.3	25.1	19.7	15.3
		MCWB	6.0	9.1	10.3	13.7	16.8	20.5	22.2	22.7	21.8	17.8	13.0	9.7
	2%	DB	9.0	11.5	15.6	20.7	24.4	26.5	30.4	30.4	27.5	22.5	17.5	12.4
		MCWB	4.5	6.9	8.6	12.3	15.8	19.5	22.2	22.6	20.8	17.1	11.8	7.7
	5%	DB	7.0	9.0	13.3	18.6	22.5	24.8	29.1	29.2	25.8	20.5	15.6	10.5
		MCWB	2.4	4.5	7.7	11.3	14.7	18.6	21.9	22.3	20.1	15.7	10.6	6.1
	10%	DB	5.3	6.9	11.3	16.7	20.8	23.1	27.5	27.9	24.2	18.9	13.8	8.6
		MCWB	1.0	2.8	6.7	10.6	14.3	17.9	21.6	21.9	19.4	14.9	9.7	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	10.5	12.8	16.6	19.4	22.2	24.1	24.3	22.8	19.7	15.2	12.1
		MCDB	10.1	12.9	15.7	19.4	23.4	26.7	29.8	29.6	27.3	23.1	17.5	13.5
	2%	WB	5.2	7.8	10.5	14.6	17.7	20.8	23.2	23.5	21.7	18.2	13.4	9.1
		MCDB	7.7	11.3	13.4	18.6	21.6	24.8	28.2	28.5	25.7	21.2	15.7	11.0
	5%	WB	2.9	5.2	9.0	13.0	16.5	19.9	22.6	23.0	20.9	16.9	11.9	6.9
		MCDB	6.1	7.9	12.1	16.5	20.4	23.1	27.2	27.4	24.3	19.5	14.1	9.3
	10%	WB	1.4	3.3	7.2	11.7	15.3	19.0	21.9	22.4	20.2	15.6	10.6	5.1
		MCDB	4.6	6.0	10.6	15.3	19.3	21.9	26.1	26.4	23.3	18.1	13.1	7.8
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.4	10.8	11.4	10.7	8.5	8.4	8.7	8.2	8.8	10.5	10.5
		MCDBR	12.4	13.3	14.2	14.1	13.7	11.2	10.9	10.7	10.4	10.8	12.3	12.6
		MCWBR	8.3	9.1	8.9	7.7	6.9	5.2	4.4	4.4	4.9	5.9	7.4	8.4
	5% WB	MCDBR	11.7	11.9	12.5	11.7	11.3	8.9	9.9	9.8	9.0	9.2	10.5	10.9
		MCWBR	8.6	9.0	9.3	7.1	6.5	4.5	4.3	4.2	4.5	5.5	7.3	8.3
Clear-Sky Solar Irradiance	taub	0.330	0.374	0.430	0.481	0.506	0.539	0.528	0.529	0.467	0.430	0.368	0.328	
	taud	2.335	2.201	2.054	1.961	1.948	1.928	2.025	2.011	2.150	2.177	2.302	2.378	
	Ebn at Noon	859	854	833	809	794	765	771	760	790	783	806	838	
	Edh at Noon	97	124	157	182	187	190	172	171	142	126	100	88	
All-Sky Solar Radiation	RadAvg	2.86	3.44	4.21	4.83	5.15	4.49	4.73	4.77	3.84	3.17	2.67	2.50	
	RadStd	0.16	0.27	0.27	0.48	0.53	0.38	0.70	0.42	0.35	0.33	0.23	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page